TANVIR ISLAM

Industrial & Production Engineering · Supply Chain & Procurement · Lean & Continuous Improvement
Rampura, Dhaka, Bangladesh
+880 1813-012320 · tanvirislam.ie@gmail.com
linkedin.com/in/tanvir-islam-7805a4305



PROFESSIONAL SUMMARY

Final-year Industrial and Production Engineering student at Ahsanullah University of Science and Technology, graduating November 2026. Trained to read a process in terms of its losses — where it gives up time, material or machine capacity, and what it takes to stop that. Applied Kaizen and root-cause analysis on a live cement plant floor, identifying 5 operational losses across a 5-packer line. Also skilled in supply chain risk analysis, sourcing and procurement planning. Seeking an entry-level role in supply chain, procurement, production or process improvement.

THESIS & RESEARCH

Undergraduate Thesis (In Progress) — Hybrid Auxetic Interference Screw — *3D-printed implant for ACL reconstruction*

- Designed a hybrid auxetic unit-cell geometry to improve fixation strength and reduce stress shielding.
- Tested unit-cell behaviour under load in ANSYS, extracting Poisson's ratio by finite element analysis.
- Evaluated 3 candidate biocomposites — PLA, PCL and HA — for additive manufacturing suitability.
- Screened 68 studies under the PRISMA 2020 protocol and built bibliometric keyword maps in VOSviewer.
- Prototyped the screw geometry on Bambu Lab hardware via Bambu Studio, tuning print settings for dimensional accuracy.

PROJECTS

Chokepoint — Supply Risk Control Tower (Personal Project, 2026) — *Monte Carlo · AHP · Linear Programming · JavaScript*

- Mapped 13 suppliers, 5 critical imports ($91M annual spend) and 5 crisis scenarios (Hormuz, Red Sea, Venezuela, Taiwan, Chattogram cyclone) for a case appliance manufacturer in Bangladesh.
- Simulated 1,000 six-month disruption runs across 4 sourcing policies; dual sourcing plus a 10-day buffer **cut expected disruption cost by 88%** and lifted worst-case fill rate from 65% to 99%.
- Scored suppliers with AHP (CR 0.015) and time-to-survive vs time-to-recover analysis, finding **51% of spend crossing a geopolitical chokepoint** and 5 of 10 supply nodes at risk of running dry.
- Modelled the 2026 DRAM price surge, where a banded long-term agreement **halved worst-case memory cost** vs spot buying; added a sourcing LP, an 8-indicator early-warning system and root-cause analysis.

Academic Projects

- Designed the workshop floor layout for the Machine Shop Laboratory in AutoCAD, sequencing machine placement across the process flow.
- Developed an adjustable ergonomic drawing table model in SolidWorks.
- Built a low-cost oxygen concentrator mask as an emergency-response prototype.

EDUCATION

B.Sc. in Industrial and Production Engineering | 2022 – Nov 2026 (Expected)
Ahsanullah University of Science and Technology, Dhaka

HSC, 2020: Hajera-Taju Degree College, Chattogram | Chattogram Board | GPA 5.00 / 5.00

SSC, 2018: Bayezid Bostami Cantonment Board High School, Chattogram | Chattogram Board | GPA 5.00 / 5.00

INDUSTRIAL EXPERIENCE

Industrial Attachment — Akij Cement Company Ltd. | June 2026

- Mapped the full production cycle of a 5-packer automated plant across 4 stages: raw material intake, VRM grinding, clinker production and CCR-monitored packaging.
- Identified 5 operational losses, including idle packer capacity, an absent belt conveyor at VRM 1 restricting material flow, and a dust hazard at Packer 3.
- Assessed the plant against Kaizen and Lean OpEx principles, tracing each of the 5 losses to a root cause and recommending a corrective action.
- Documented 3 operational strengths (fly-ash-free production, 12-step quality control, closed-loop recycling) and 5 improvement recommendations in a plant visit report.

SKILLS

Supply Chain & Procurement: Supply Risk Assessment · Strategic Sourcing · Inventory Policy & Safety Stock · Procurement & Price-Risk Strategy · Monte Carlo Simulation · AHP (MCDM) · Linear Programming

Software & Tools: MS Excel · AutoCAD · SolidWorks · ANSYS (FEA) · MATLAB · Minitab · JavaScript (HTML/CSS) · Bambu Studio · VOSviewer · MS Word, PowerPoint

Coursework & Exposure: Supply Chain & Inventory Management · Lean Manufacturing · Kaizen & Continuous Improvement · Root-Cause Analysis · Work Study (Time & Motion) · Production Planning & Operations · Quality Control & Assurance

Certification & Training: Lean Six Sigma White Belt (GLSS, 2026) · Supply Chain & Procurement (ISCEA Ptak Prize Competition)

Core Competencies: Problem Solving · Analytical Thinking · Leadership · Communication · Teamwork

Languages: Bengali — Native · English — Fluent · Hindi — Conversational

ACHIEVEMENTS

- 3rd Place, IPE Product Showcasing — AUST, 2025 · Winner, Physics & Mathematics Olympiad

REFERENCES

Mr. Zahid Hasan Shuvo — Thesis Supervisor, Asst. Professor, IPE, AUST · zhshuvo.ipe@aust.edu · +880 1753-575669

Sohanur Rahman Sohan — Deputy COO, Akij Resource · deputy.coo@akijresource.com · +880 1712-684048